

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Third Semester B.E. (E&E) (Credit System) Degree Examinations**

April 2023

21EE303 – DIGITAL SYSTEM DESIGN

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1. a)	Determine prime implicants, essential prime implicants and minimal sum for the given function $F(a,b,c,d)=\sum(0,1,2,8,5,15)+\sum d(6,7,10)$ using Karnaugh Map technique.	8	L*3	1	1
b)	Find the minimal sum using MEV technique for $F(w,x,y,z)=\sum(2,3,4,5,13,15)+\sum d(8,9,10,11)$ considering w,x,y as map variables.	6	L3	2	1
c)	Explain the carry look-ahead adder principle. With the help of neat diagram derive logic expressions for the carry outputs of a 4-stage carry look -ahead adder.	6	L2	2	1
2. a)	Draw the logic diagram for 3to8 decoder having active low data inputs, an ative high enable and active low data outputs. Using that realize Boolean equation given $F(a,b,c)=\sum(0,3,5,6)$	7	L3	2	1
b)	Realize the following Boolean function using appropriate multiplexer whose data inputs are to be connected directly to logic levels 0 and 1. $f(a,b,c,d)=\prod(0,2,3,6,9,10,11,14)$	7	L3	2	1
c)	Use K-map to simplify the following: $f(a,b,c,d)=\sum(2,3,4,5,13,15)+\sum d(8,9,10,11)$	6	L2	1	1
3. a)	Design a combinational circuit for a two-bit magnitude comparator.	8	L3	2	1
b)	Implement $f(a,b,c,d)=\sum m(1,4,5,7)$ using 4 to 1 MUX using a and b as the address lines.	8	L3	2	1
c)	Simplify using K-map: $T=w'xy+wz'+xyz'$	4	L3	1	1
Unit – II					
4. a)	Draw a neat circuit diagram of Master -Slave JK flipflop. Explain 0's and 1's catching problem with the help of timing diagram.	10	L2	3	1
b)	Write the logic diagram, truth table and timing diagram of gated D latch. Also obtain the characteristic equation.	10	L2	3	1
5. a)	Explain the working of switch denouncer using SR latch with the help of diagram.	6	L2	3	1
b)	Design a Mod-6 counter to count the sequence 0,2,3,6,5,1,0 using positive edge triggered JK flip-flops by obtaining its minimal sum equations.	8	L3	4	1
c)	With a neat logic diagram explain the following i) SIPO shift register using D flipflop ii) Johnson counter	6	L2	4	1

6. a) Derive the characteristic equations for the following flipflops by constructing appropriate K-maps and obtaining minimal sums.

i) T Flipflop

ii) SR flipflop

10 L2 3 1

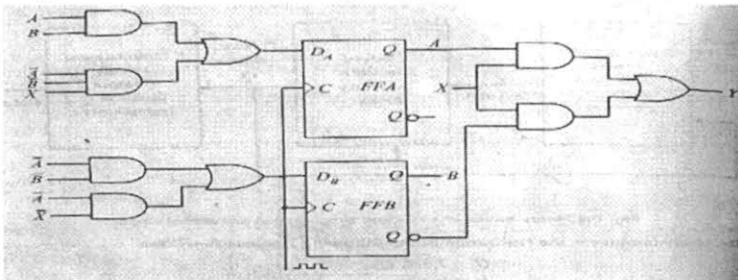
- b) Design a Universal shift register incorporating four multiplexers and four positive edge triggered D flipflops having states specified in the table below.

Select lines		Register operations
S1	S0	
0	0	Hold
0	1	Synchronous clear
1	0	Complement contents
1	1	Circular shift right

10 L3 4 1

Unit – III

7. a) Construct the excitation table, transition table, state table and state diagram for the sequential circuit below.

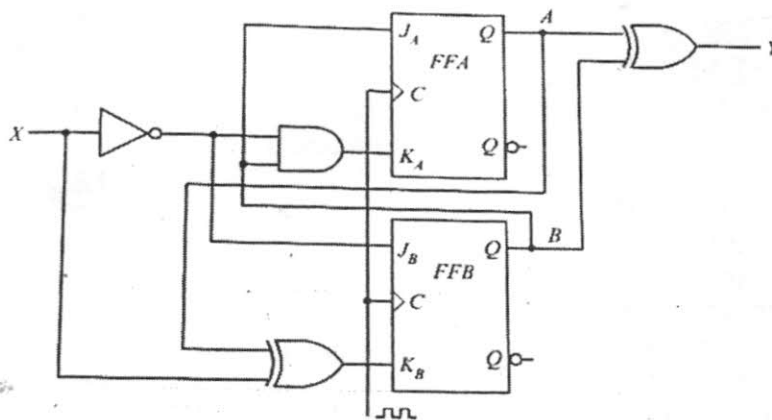


10 L3 5 1

- b) Discuss the operations of CMOS inverter and CMOS NAND Gates.

10 L2 5 1

8. a) Construct the excitation table, transition table, state table and state diagram for the sequential circuit below.



10 L3 5 1

- b) Discuss the operation of NMOS NOT, NAND and NOR gates.

10 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
